

Multiple Choice

1. **Answer:** B

Options: A) 1 meter; B) 25 meters; C) 1 kilometer; D) 25 kilometers

Note: Correct option: B - 25 meters

2. **Answer:** D

Options: A) 10; B) 45; C) 125; D) 25

Note: Correct option: D - 25

3. **Answer:** B

Options: A) -4; B) 0; C) 2; D) 4

Note: Correct option: B - 0

4. **Answer:** D

Options: A) 12; B) 3; C) 5; D) 6

Note: Correct option: D - 6

5. **Answer:** C

Options: A) $(-4, 4)$; B) $[0, 4]$; C) $(-\infty, -1) \cup (4, \infty)$; D) $(-\infty, -1) \cup (-1, 4) \cup (4, \infty)$

Note: Correct option: C - $(-\infty, -1) \cup (4, \infty)$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '42'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '8.1'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '36'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** For $(ax + b)(x^5 + 1) - (5x + 1)$ to be divisible by $x^2 + 1$, it must be equal to 0 at the roots of $x^2 + 1 = 0$, which are $\pm i$. For $x = i$, $(ax + b)(x^5 + 1) - (5x + 1)$
=&= $(ai + b)(i + 1) - (5i + 1)$
=&= $-a + ai + bi + b - 5i - 1$
=&= $(-a + b - 1) + (a + b - 5)i$. Then we must have $-a + b - 1 = a + b - 5 = 0$. Solving, we find $(a, b) = (2, 3)$. For these values,

$$(ax + b)(x^5 + 1) - (5x + 1) = 2x^6 + 3x^5 - 3x + 2 = (x^2 + 1)(2x^4 + 3x^3 - 2x^2 - 3x + 2).$$

Note: Students should show all steps in their mathematical solution.

12. **Answer:** $200 + 450 + 200$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key